OPEN-SOURCE CLINICAL DECISION SUPPORT

PRAHARI

A digital twin per bed, for the cardiac step-down ward at night.

प्रहरी — the one who keeps watch



Madhav Sharma

Digital twin for heart failure

THE WARD WE ARE BUILDING FOR

One doctor. Eight failing hearts. A chart every four hours.

1 : 75

intensivists to ICU beds in India — and the wards below the ICU get less

4 h

between charted vitals on a typical step-down ward; no continuous telemetry recorded

30.8%

one-year mortality after an acute heart-failure admission, Trivandrum HF Registry

Heart-failure patients deteriorate over hours, not seconds — slowly enough to act on, quietly enough to miss between rounds.

Sources: ISCCM · Trivandrum Heart Failure Registry

WHAT THE WARD ALREADY HAS

NEWS2 says who is sick. It can't say what kind — and in heart failure, that decides the drug.

NEWS2 11 | C · COLD & WET

Congested **and** underperfused.

Needs inotrope + diuretic, and escalation.

NEWS2 5 | L · COLD & DRY

Underperfused, **not** congested.

The reflex diuretic makes her worse. Hold it; cautious fluid.

Both alarms fire. The right response is opposite. The ward has no instrument that tells them apart at 2 a.m.

THE IDEA

A runnable copy of each patient's heart, fitted from the chart the nurse already keeps.

1

State. Contractility, resistance, volume, lung water — fitted to *this* patient.

2

Forward. Simulates the next 12 hours as a trajectory, not a score.

3

Forks. Give the drug now, in four hours, or not at all — three futures.

4

Sync. Re-fits at every reading; keeps running between them.

The test of a twin: **can you ask it about a future that never happened?** A risk score can't.

PRAHARI

a digital twin per bed, for the cardiac step-down ward

ARCHITECTURE

1 · WHAT THE WARD HAS

STATIC · EHR

Admission record

Age, sex, body surface area
Diagnosis, aetiology, LVEF, NYHA
BNP, creatinine, Na, K, Hb, HbA1c
Home medication
CYP2D6 genotype → drug response
sets priors and patient constants

DYNAMIC · BEDSIDE MONITOR

Charted every 4 hours

HR · BP · SpO_2 · resp. rate (monitor)
JVP · oedema · weight (ward round)
→ identify congestion (wet / dry)
Capillary refill time
→ identifies perfusion (warm / cold)
Noisy, rounded, ~10% of rounds missed
no networked telemetry, no catheter

AT HOME · WEARABLES

A ₹2,000 band + a scale

Resting HR · HRV · sleep · steps · weight
read from an Apple Health export
same heart → days to decompensation

2 · ASSIMILATE

Learned inverse

Gradient boosting trained on
6,000 simulated physiologies —
no patient data needed
→ fast first guess

Joint trajectory fit

Every reading fitted at once,
weighted by its measurement noise,
older readings weighted less
bounded least squares + priors

Out

Contractility, resistance, volume,
lung water, arterial compliance
+ trends: is the heart failing?
+ confidence: can it say
"I don't know"?

3 · THE TWIN

PHYSICS · NOT A BLACK BOX

Lumped-parameter heart

$SV = Ees(EDV - V_0)/(Ees + Ea)$

Ventricular–arterial coupling
Saturating Frank–Starling curve
Congestion penalty on contractility
Baroreflex on heart rate and SVR
Cardiorenal fluid balance
Pulmonary filtration → SpO_2

A runnable copy of one patient

① state fitted to this patient
② simulates forward in time
③ forks under any decision
④ re-syncs at every reading, and
keeps running between them

reads out as a Stevenson profile:
wet/dry × warm/cold

4 · FORK AND DECIDE

COUNTERFACTUAL ENGINE

Every drug is a parameter lever

Diuretic → volume · nitrate → SVR, preload
Inotrope → contractility · pressor → SVR
Fluids → volume · β-blocker → HR (×CYP2D6)
Onset, plateau and washout per agent

Act now · act in N hours · wait

forked from the patient as they are now

DECISION LAYER

Behind the early-warning score

NEWS2 already says who is sick.
The twin says what kind of sick —
warm-wet, cold-wet, cold-dry — and
what acting or waiting will do.

Deadline to cold-wet · escalation rule
Ranks the ward: who to see first
advises; never issues orders

CLINICIAN STATION

Ward view — every bed, live traces, NEWS2 beside the twin's reading, ranked

Patient view · haemodynamic map · wait-or-act · trends with forecast band · record · handover brief · the twin at home

FastAPI + hand-written browser client · runs on any laptop · no GPU, no cloud, no patient data needed to run

HOW WE KNOW — SIMULATED GROUND TRUTH, THEN REAL DATA

Simulated ward
Registry-shaped patients,
true physiology known

→

Ward degrader
4-hourly, noisy, rounded,
10% of rounds missed

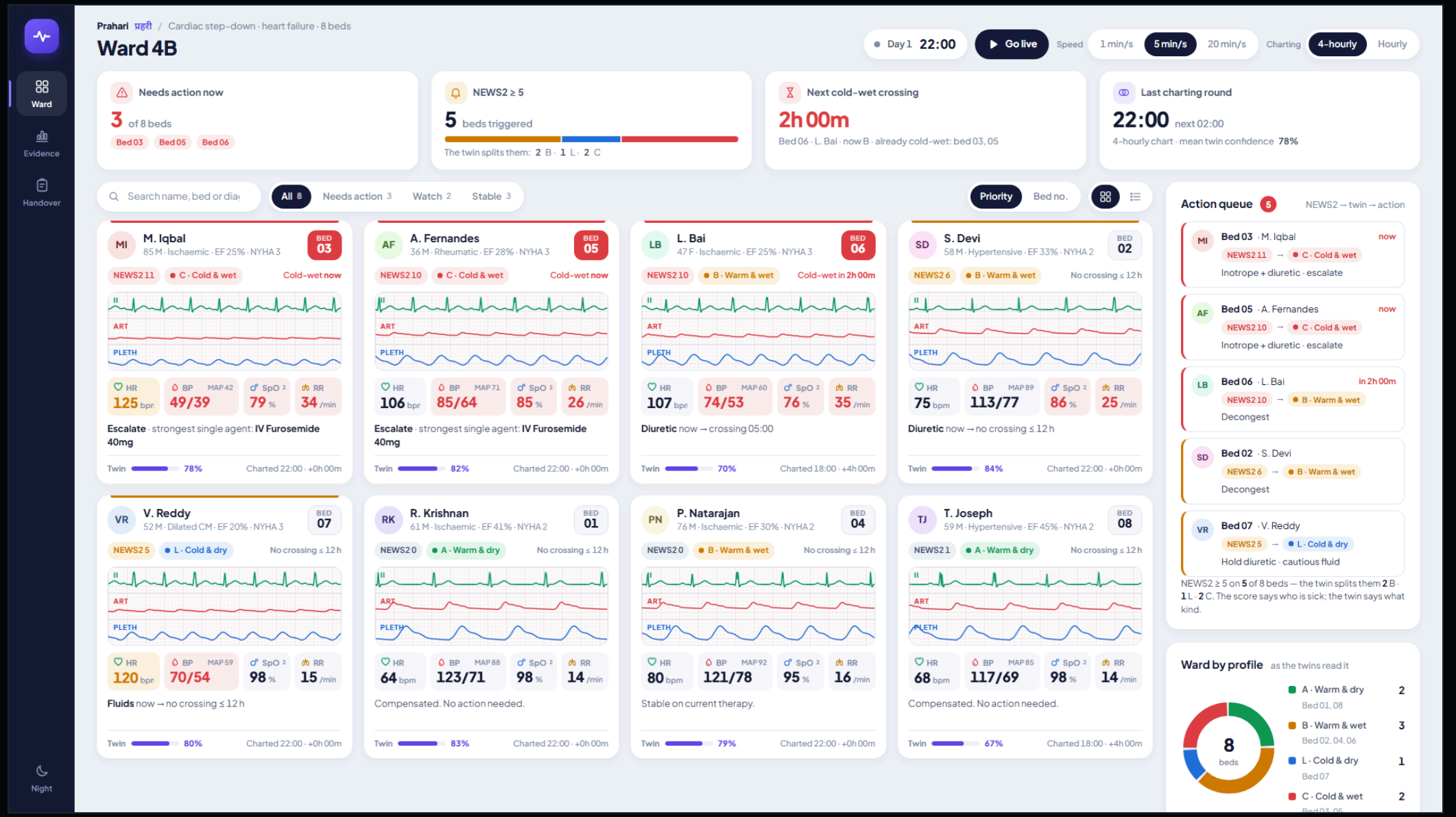
→

Twin sees
only the chart —
never the truth

→

Scored against the truth
① hidden state at 4-hourly vs hourly charting
② every drug given to twin and to "real" patient
③ round-by-round replay and alarm curves vs NEWS2

Then on real data: 32 heart-failure ICU stays from MIMIC-IV (CVP, cardiac output and PA pressure withheld as answer keys) · 11 Fitbit volunteers read through Apple Health exports



Every bed: the twin's virtual patient, NEWS2 beside the twin's reading, ranked

THE QUESTION AT THE END OF A SHIFT

"Can this wait until the morning round?"

DO NOTHING

12.3 h

cold-wet in the next 14 h

DIURETIC IN 4 H

12.3 h

she crosses before it lands

DIURETIC NOW

7.3 h

crossing pushed back

> Bed 6's 22:00 reading was missed. The twin kept simulating, dropped its confidence, and projected cold-wet at midnight. Act before you leave.

EVIDENCE 1 · HIDDEN STATE FROM A 4-HOURLY CHART

The twin estimates what the ward can't measure — without continuous monitoring.

Charting	Profile correct	Wet vs dry	Cold vs warm	Cardiac index error	Filling pressure error
Hourly	96%	98%	98%	0.17	0.9 mmHg
4-hourly, by hand	83%	91%	93%	0.28	1.6 mmHg

Same 96 patient-states in both rows. The fix that made it work was clinical, not mathematical: a BP cuff can't see filling pressure, so the twin reads the signs doctors already chart — JVP, oedema, weight for **wet**; capillary refill for **cold**.

EVIDENCE 2 · ARE THE WHAT-IFS RIGHT?

Every drug, given to the twin and to the true patient at the same moment.

99%

drug effects predicted in the right direction (3,080 comparisons)

r 0.97

predicted vs true change in cardiac index

9×

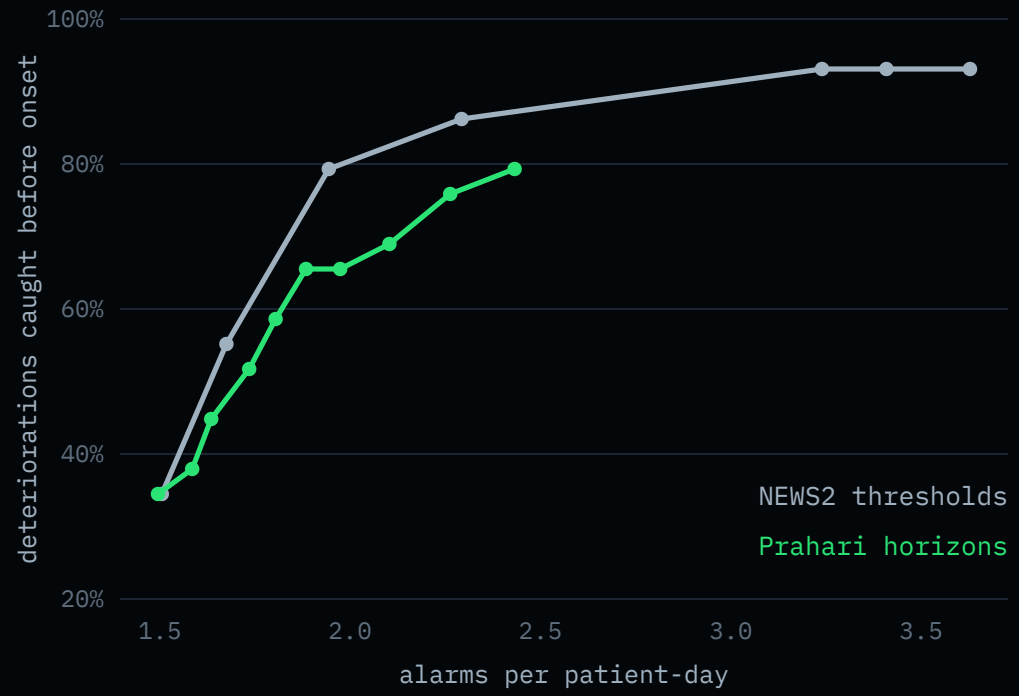
less decision regret than "give furosemide whenever NEWS2 fires"

True profile	Furosemide helps	Furosemide harms	Twin names the profile
A — warm, dry	0%	100%	84%
B — warm, wet	72%	28%	85%
L — cold, dry	0%	100%	91%
C — cold, wet	79%	0%	92%

Checked against simulated patients' pharmacology · real-dose calibration pending (slide 11)

EVIDENCE 3 · WHERE WE ARE NOT BETTER

As an alarm, the twin is no better than NEWS2. We measured it.



Both systems replayed round by round on 29 deteriorations, thresholds swept. At matched alarm burden, NEWS2 catches as many or more.

With one or two readings, a downward trend isn't knowable yet — the twin misses some early slides.

So Prahari sits behind the score. NEWS2 says who. The twin says what kind, what to do, and what waiting costs.

We withdrew an earlier "81% fewer alarms" claim when a fair comparison didn't support it

EVIDENCE 4 · REAL PATIENTS, REAL WEAR DATA

We tested it where it could fail — and it did, in useful ways.

MIMIC-IV · 32 REAL HEART-FAILURE ICU STAYS

Twin vs withheld measurement	Stays	r
Central venous pressure	10	0.30
Cardiac index	3	0.62
Wedge vs PA diastolic	3	0.53

Moves with real measurements, but reads patients as less sick than they are — the unsafe direction. 4-hour vital-sign forecasts don't beat "no change" (10.2 vs 9.3 mmHg).

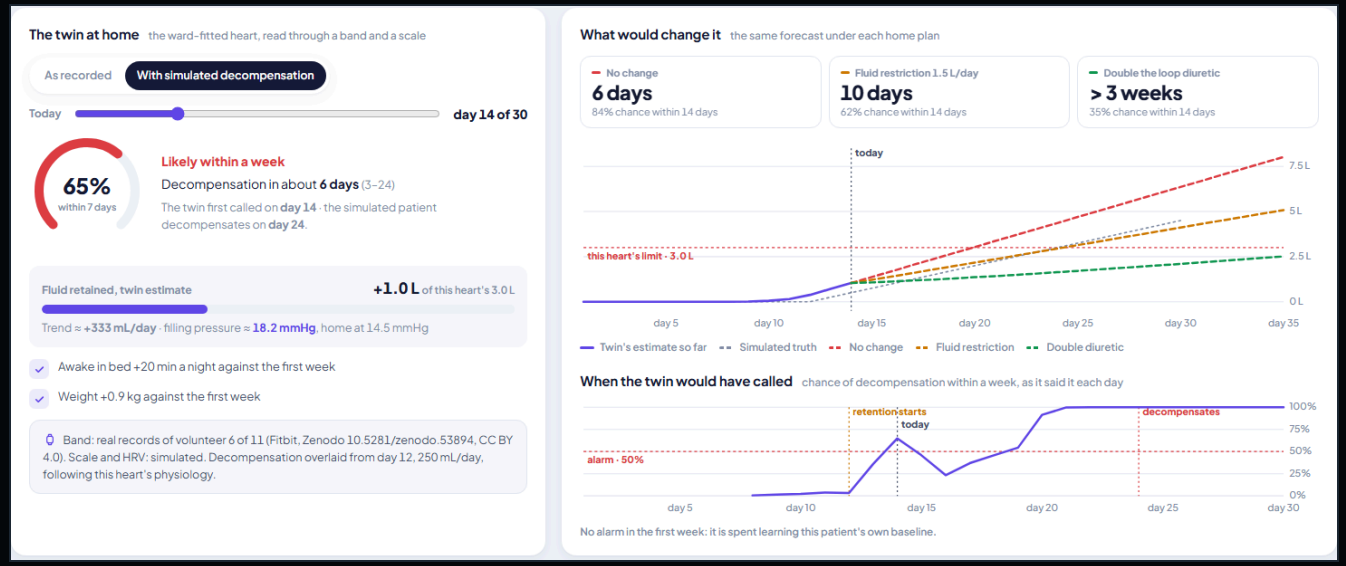
AT HOME · 96 PATIENTS ON REAL FITBIT NOISE

Detector	Caught ahead	Days
"2 kg in 3 days" rule	17 of 91	4.0
"2 kg over first week"	84 of 91	4.5
Home twin, band + scale	91 of 91	6.0
Home twin, band only	61 of 91	4.0

On **real** volunteers' records: 0.6% false alarms with a scale, 3.9% on the band alone — **too many to use the band by itself**.

Small samples, reported as they came out. The next step is the full MIMIC-IV database with a proper train/test split.

The same heart goes home — and says how many days it has left.



A week of band and scale data, read by the ward-fitted heart: resting heart rate, HRV, sleep, time awake in bed, steps, weight.

Each heart has its own limit — 2.2–3.8 litres retained. Weak hearts go home fuller and tip over sooner.

Forecast at the first alarm off by 2.2 days; the true day inside its range 89% of the time.

Wait or act, in days: fluid restriction or a doubled diuretic, run on the same heart.

BUILT FOR HERE

Designed around what an Indian ward actually has.

No new hardware. Fitted from the 4-hourly chart and bedside signs — no telemetry, no catheter.

The twin goes home. A ₹2,000 band and a bathroom scale — no clinic visit — and it counts the days to decompensation.

Registry-shaped patients. Younger, sicker, rheumatic disease still present, heavy diabetes overlap.

Pharmacogenomics. CYP2D6 phenotype changes the simulated beta-blocker response.

Food is physiology. A rice-heavy dinner drives glucose and sympathetic load in diabetic heart failure; millet cuts both.

Runs on a laptop. No GPU, no cloud, no patient data needed to train.

What this is, and what it is not.

Not clinically validated. The simulated results are an upper bound; on real ICU patients the twin is miscalibrated towards "less sick" and its drug effect sizes are uncalibrated.

Not a better alarm. It sits behind NEWS2. At home the band alone false-alarms too often; with a scale it matches the best weight rule.

Not an order-giver. It advises and explains; the doctor decides.

Next: the full, credentialed MIMIC-IV — recalibrate on a training split, test on the rest.

Then: silent shadow mode on a ward — the twin predicts, nobody acts on it, we compare.

Model: infusion titration and repeat dosing; right-heart physiology.

PRAHARI

Who to see first. What kind of sick. What waiting will cost.

github.com/Madhav0414/prahari



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